

# Minseong Cho

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🏢 Softlab, Ajou University  
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📁 Portfolio   
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## Research Interests

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Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

## Education

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- **M.S. in Mechanical Engineering** Sep 2025 – Present  
Ajou University, Suwon, South Korea
  - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025  
Ajou University, Suwon, South Korea

## Research Experience

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- **Graduate Research Student** Sep 2025 – Present  
Softlab, Ajou University
  - **Mechanics of Moisture-Driven Hair Frizz**
    - **Focus:** Moisture-driven changes in strand morphology.
    - **Techniques:** Centerline-based geometric analysis and simulation-assisted interpretation.
    - **Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
  - **Boundary-Driven Active Matter in Soft Hydrogels**
    - **Focus:** Leidenfrost-induced dynamics in soft hydrogel systems.
    - **Techniques:** Image-based tracking, curvature analysis, and visualization scripts.
    - **Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student** Mar 2023 – Aug 2025  
Softlab, Ajou University
  - **Boundary-Driven Active Matter in Soft Hydrogels**
    - **Focus:** Leidenfrost-related soft-matter dynamics.
    - **Techniques:** Image-based tracking, quantitative analysis, and figure preparation.
    - **Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung

## Conferences and Presentations

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| <b>APS Division of Fluid Dynamics Meeting</b> (Houston, TX, USA)<br>Leidenfrost-induced active matter dynamics of spherical hydrogels | Nov 2025 |
| <b>Asian Congress of Fluid Mechanics</b> (Seoul, South Korea)<br>Leidenfrost-induced active matter dynamics of spherical hydrogels    | Sep 2025 |
| <b>KSME Annual Conference</b> (Jeju, South Korea)<br>Explosive motion of spherical hydrogel via the Leidenfrost effect                | Nov 2024 |

## Awards & Honors

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- **Dean's List** Spring 2024, Spring 2025  
Ajou University  
Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** June 2025  
Ajou University  
**2nd Place · KRW 1,000,000 prize**
  - **Project:** Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025  
**3rd Place · KRW 300,000 prize**
  - **Organized by:** Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)
  - **Project:** Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** 2025  
**US\$750 travel cost reimbursement**
  - **Organized by:** APS Division of Fluid Dynamics, American Physical Society
  - **Purpose:** Travel support for participation in the APS Division of Fluid Dynamics Meeting.
- **Gallery of Fluid Motion** Apr 2026  
**Haechon Choi GFM Award · KRW 400,000 prize**
  - **Organized by:** The Korean Society of Mechanical Engineers, Fluid Engineering Division
  - **Title:** Explosive motion of spherical hydrogel via the Leidenfrost effect
  - **Authors:** **Minseong Cho**, Yeonsu Jung, Jonghyun Ha

## Publications

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### Boundary-driven active matter and stochastic rectification via Leidenfrost impulses

Minseong Cho, Yeonsu Jung, and Jonghyun Ha

✉ In preparation

### Poroelastic swelling time as a quantitative marker of hair damage and treatment

Jaehyuk Cho, **Minseong Cho**, Selin Cho, Haesoo Park, Hyun-Na Kim, and Jonghyun Ha

✉ In preparation

## References

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### Prof. Jonghyun Ha

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### Dr. Yeonsu Jung

Postdoctoral Research Fellow, School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA

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